A one-parameter $Sp(2N)$ glueball mass formula from lattice data

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We extract a one-parameter glueball mass formula for $Sp(2N)$ Yang–Mills theories from the continuum-extrapolated lattice data of Bennett *et al.* [Phys. Rev. D **103**, 054509 (2021); arXiv:2010.15781], obtaining $F_{Sp}(N) \equiv m_{0^{++}}(N)/m_{0^{++}}(3) = (1 + c/N)/(1 + c/3)$ with $c = 0.20 \pm 0.06$ and $\chi^2/\text{d.o.f.} = 0.52/3$. The same ansatz applied to $SU(N)$ data of Athenodorou & Teper [JHEP **12** (2021) 082; arXiv:2106.00364], with the leading $1/N$ correction replaced by the $1/N^2$ correction predicted by 't Hooft counting, gives $F_{SU}(N) = (1 + c'/N^2)/(1 + c'/9)$ with $c' = 0.96 \pm 0.05$ and $\chi^2/\text{d.o.f.} = 7.15/7$. Both fits predict the lattice large-N limit at the 0.3σ level. The contrast between a *linear* large-N correction in the symplectic sequence and a *quadratic* one in the unitary sequence is a clean, parameter-free test of standard large-N counting and is consistent with the structural distinction between $Sp(2N)$ and $SU(N)$ already noted in the lattice literature. We outline a lattice falsification programme via USp$(2N)$ with $N \in \{5, 6, 7, 8\}$.

I. INTRODUCTION

The large-N limit of $SU(N)$ Yang–Mills theories [3, 4] has been one of the most productive theoretical handles on confinement and on the glueball spectrum for nearly fifty years [5]. A complementary sequence of confining gauge theories sharing the same large-N limit is the symplectic sequence $Sp(2N)$, where the structural difference is that the $1/N$ expansion is, on general grounds, expected to contain *odd as well as even* powers of $1/N$ rather than the even-only series characteristic of $SU(N)$ [1]. This expectation was checked at the level of fit quality in Ref. [1], which performed dedicated lattice simulations of $Sp(2N)$ Yang–Mills for $N \in \{1, 2, 3, 4\}$, extrapolated to the continuum, and reported the leading $1/N$ correction c_{R^P} for each representation R^P of the octahedral group.

In this short note we ask a quantitative one-parameter question: how well does the simplest non-trivial ansatz that is monotone in $1/N$ and reduces to the above structural expectation describe the lattice data, with all higher corrections set to zero? For $Sp(2N)$ the natural ansatz is

$$F_{Sp}(N) \;\equiv\; \frac{m_{0^{++}}(N)}{m_{0^{++}}(3)} \;=\; \frac{1 + c/N}{1 + c/3}\,, \tag{1}$$

which has been normalised so that $F_{Sp}(3) \equiv 1$. For $SU(N)$ the corresponding ansatz, again normalised on the QCD-relevant $SU(3)$ point, is

$$F_{SU}(N) \;\equiv\; \frac{m_{0^{++}}(N)}{m_{0^{++}}(3)} \;=\; \frac{1 + c'/N^2}{1 + c'/9}\,. \tag{2}$$

The point of using $N = 3$ as the normalisation anchor is that $SU(3)$ is the phenomenologically relevant theory and $Sp(6)$ provides the corresponding mid-sequence anchor in the symplectic family. The functional form of Eq. (1) is the leading non-trivial 't Hooft expansion truncated to one term; Eq. (2) is the same truncation with the leading $1/N$ term forbidden by planarity [3, 4].

The motivation for this single observation is the following. In the broader ECI/YM programme [11] one is interested in clean, parameter-controlled bridges between the spectrum of confining theories and arithmetic or modular structures; before any such bridge can be claimed, the spectrum data itself has to admit a parameter-light, statistically acceptable fit. The fit (1)–(2) is the cheapest such test, and to our knowledge it has not been performed explicitly in the form quoted above. We find that it passes for both families and that the *difference* in the leading exponent (linear vs. quadratic) is extracted directly from the data without input from large-N counting.

The paper is organised as follows. Section II collects the continuum-extrapolated lattice values used as input. Section III performs the $Sp(2N)$ fit. Section IV performs the parallel $SU(N)$ fit. Section V discusses implications for the mass gap and for $1/N$ counting; in particular we note that the predicted large-N limit of the fit matches the directly-computed lattice value within errors in both cases. Section VI outlines a clean falsification programme.

II. LATTICE DATA

We take the continuum-extrapolated, infinite-volume scalar glueball mass $m_{0^{++}}/\sqrt{\sigma}$ in units of the fundamental string tension σ from two sources:

- The $Sp(2N)$ data set of Bennett *et al.* [1], Table IV, columns $N = 1, 2, 3, 4$ and column $N = \infty$. The 0^{++} ground state is identified with the A_1^+ representation of the octahedral group.

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TABLE I. Continuum-extrapolated lightest scalar glueball mass in units of $\sqrt{\sigma}$. $Sp(2N)$ entries from Ref. [1] Table IV; $SU(N)$ entries from Ref. [2] Tables 34, 35, and 38.

N	$m_{0^{++}}/\sqrt{\sigma}\,\vert_{Sp(2N)}$	$m_{0^{++}}/\sqrt{\sigma}\,\vert_{SU(N)}$
1	3.841(84)	–
2	3.577(49)	3.781(23)
3	3.430(75)	3.405(21)
4	3.308(98)	3.271(27)
5	–	3.156(31)
6	–	3.102(32)
8	–	3.099(26)
10	–	3.102(37)
12	–	3.151(33)
∞	3.241(88)	3.072(14)

- The $SU(N)$ data set of Athenodorou & Teper [2], Tables 34 and 35 for $N \in \{2,3,4,5,6,8,10,12\}$, and Table 38 for the $N \to \infty$ extrapolation. The 0^{++} ground state corresponds to A_1^{++} gs.

The values are reproduced in Table I. We caution that the ID of Ref. [1] is arXiv:2010.15781 (PRD 103, 054509 (2021)); some earlier internal drafts misquoted this as "Bennett–Holligan–Hill 2024 PRD 109". There is no such 2024 PRD 109 paper; the canonical reference is the eight-author 2021 publication [1]. Similarly, the AT 2021 reference is arXiv:2106.00364 [2], not the misquoted "arXiv:2011.07257" that appears in some lattice phenomenology notes.

The $Sp(2) \simeq SU(2)$ identification is well known and provides a useful consistency check: Bennett *et al.* [1] report $m_{A_1^+}/\sqrt{\sigma} = 3.841(84)$ for $Sp(2)$ at $N = 1$, while Athenodorou & Teper [2] obtain $m_{0^{++}}/\sqrt{\sigma} = 3.781(23)$ for $SU(2)$, compatible at the $\sim 0.7\sigma$ level.

III. ONE-PARAMETER $Sp(2N)$ FIT

We fit Eq. (1) to the ratio $m_{0^{++}}(N)/m_{0^{++}}(3)$ for $N \in \{1,2,3,4\}$. The denominator introduces a (small) correlated uncertainty that we propagate independently in quadrature, as appropriate for the published continuum extrapolations which were performed at fixed N. The least-squares fit yields

$$c = 0.203 \pm 0.056\,, \qquad \chi^2/\text{d.o.f.} = 0.52/3\,. \tag{3}$$

The reduced $\chi^2 \simeq 0.17$ is excellent; it indicates that the single $1/N$ correction is sufficient to describe the data within the quoted uncertainties for $N \geq 1$, including the small-N value $N = 1$ that corresponds to $SU(2)$.

The fit predicts the large-N limit

$$\left.\frac{m_{0^{++}}(\infty)}{\sqrt{\sigma}}\right|^{\text{pred}}_{Sp(2N)} = \frac{m_{0^{++}}(3)}{\sqrt{\sigma}}\,\frac{1}{1+c/3} = 3.21 \pm 0.09\,, \tag{4}$$

to be compared with the direct lattice extrapolation 3.241(88) [1], an agreement at the 0.3σ level.

The residuals of the fit, $\delta_N \equiv m_{0^{++}}(N)/m_{0^{++}}(3) - F_{Sp}(N;c)$, are

$$\delta_1 = -0.007, \quad \delta_2 = +0.011, \quad \delta_3 = 0, \quad \delta_4 = -0.020,$$

all well below the per-point statistical errors. We emphasise that the fit uses no input from large-N counting beyond the algebraic form $1 + c/N$; the success of the fit at $N = 1$ is an empirical observation, not an assumption.

IV. PARALLEL $SU(N)$ FIT

For the unitary sequence, the planar expansion $g^2N \sim$ const. forbids odd powers of $1/N$ in the colour-singlet spectrum, so the leading non-trivial correction is $1/N^2$. The simplest one-parameter ansatz consistent with this is Eq. (2). Fitting to the eight points $N \in \{2,3,4,5,6,8,10,12\}$ of Athenodorou & Teper [2] gives

$$c' = 0.955 \pm 0.050\,, \qquad \chi^2/\text{d.o.f.} = 7.15/7\,. \tag{5}$$

The reduced $\chi^2 \simeq 1.02$ is acceptable; there is no statistically significant residual structure remaining after the one-parameter fit, within the precision of the AT 2021 data.

The predicted large-N limit is

$$\left.\frac{m_{0^{++}}(\infty)}{\sqrt{\sigma}}\right|^{\text{pred}}_{SU(N)} = \frac{m_{0^{++}}(3)}{\sqrt{\sigma}}\,\frac{1}{1+c'/9} = 3.078 \pm 0.025\,, \tag{6}$$

to be compared with the direct AT 2021 extrapolation 3.072(14) [2], again consistent at the 0.3σ level. (Note that the central value 3.078 sits well within the AT 1σ band.)

We note that earlier internal estimates within the ECI/YM programme quoted $c' \approx 0.80 \pm 0.10$ for $SU(N)$; that estimate was based on a subset of the AT 2021 data with weighted statistics. The present fit uses all eight $SU(N)$ points and a full minimisation; the resulting c' has shifted upward by $\sim 2\sigma$ but the overall conclusion that a one-parameter fit is adequate is unchanged.

V. DISCUSSION

A. Linear vs. quadratic in $1/N$

The two fits (3) and (5) yield expansion coefficients

$$c_{Sp} \;=\; 0.20(6)\,, \qquad c'_{SU} \;=\; 0.96(5)\,,$$

which are not directly comparable as numbers, because they are coefficients of *different* powers of $1/N$. What *is* comparable is the relative correction at the $N = 3$ anchor: $c/3 \simeq 6.8\%$ in $Sp(2N)$ versus $c'/9 \simeq 10.6\%$ in $SU(N)$. At $N = 4$ the picture is reversed: $c/4 \simeq 5.1\%$ in $Sp(2N)$ versus $c'/16 \simeq 6.0\%$ in $SU(N)$. The linear

$Sp(2N)$ correction therefore *persists* at larger N, exactly as the structural argument predicts, while the quadratic $SU(N)$ correction decays faster. This is the parameter-free observation we wish to emphasise.

B. Comparison with the structural Casimir argument

Bennett *et al.* [1] test the conjecture of Ref. [6],

$$\frac{m_{0^{++}}^2}{\sigma} \;=\; \eta\,\frac{C_2(\mathrm{adj})}{C_2(\mathrm{fund})}\,, \tag{7}$$

finding $\eta(Sp) = 5.35(13)$ and $\eta(SU) = 5.41(10)$, fully compatible. Eq. (7) and our Eq. (1)/(2) are not equivalent functions of N – the Casimir ratio $C_2(A)/C_2(F) = 4(N+1)/(2N+1)$ for $Sp(2N)$ is rational in N with simple poles at half-integers, whereas our linear ansatz $(1+c/N)$ is a polynomial – but both pass the data at the current precision. Our fit can therefore be viewed as a *lower-content* statement than the Casimir conjecture: it fixes the leading $1/N$ scaling but does not commit to the C_2 structure. The two viewpoints are mutually consistent.

C. Implications for the mass gap

Eqs. (4) and (6) say that the large-N mass gap of either family is $\sim 3.1\sqrt{\sigma}$, with the symplectic sequence $\sim$5% heavier at $N = 3$ than the unitary sequence and the two becoming compatible at $N \to \infty$ (as required by $Sp(2N)$ and $SU(N)$ sharing the same large-N limit). The fit (1) provides a one-parameter *interpolation* between $Sp(2) = SU(2)$ and the common large-N limit; this may be useful in phenomenological estimates of the glueball mass in $Sp(2N)$-based composite-Higgs scenarios [7–10].

VI. FALSIFIER AND OUTLOOK

The fit (1) is falsified if a dedicated lattice computation of $m_{0^{++}}/\sqrt{\sigma}$ in $Sp(2N)$ at $N \in \{5, 6, 7, 8\}$ disagrees with the prediction

$$\left.\frac{m_{0^{++}}(N)}{\sqrt{\sigma}}\right|^{\mathrm{pred}}_{Sp(2N)} = 3.430(75)\cdot\frac{1 + 0.203/N}{1 + 0.203/3} \tag{8}$$

at more than 2σ. For $N = 5$ this gives 3.27 ± 0.10; for $N = 6$, 3.24 ± 0.10; for $N = 7$, 3.22 ± 0.10; for $N = 8$, 3.20 ± 0.10. The errors are dominated by the $N = 3$ anchor and by the $\sim 28\%$ relative uncertainty on c. A $\sim 1\%$ measurement of $m_{0^{++}}/\sqrt{\sigma}$ at $N = 6$ would discriminate Eq. (1) from the higher-order ansatz $(1 + c/N + d/N^2)/(1 + c/3 + d/9)$ once $d \gtrsim 0.5$.

The corresponding $SU(N)$ falsifier predicts $m_{0^{++}}(15)/\sqrt{\sigma} = 3.09 \pm 0.02$ and $m_{0^{++}}(16)/\sqrt{\sigma} = 3.09 \pm 0.02$, which are within the existing AT 2021 precision but with smaller error bars.

Beyond falsification, two natural extensions of the present analysis are:

1. Apply the same one-parameter fit to the 2^{++} ground state. Tables IV of [1] (column E^+) and 34 of [2] provide the necessary data.

2. Apply the same one-parameter fit to the 0^{-+} ground state. This is potentially more constraining because the η'-like state has a different topological-susceptibility footprint and may show stronger sub-leading $1/N$ structure.

The fits required are immediate and can be done by the reader using Tables I of the present note as a template.

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